Core Concepts in Quantum Field Theory

YU LIU

This note is a partial reading note of Peskin's book. YL: [However, this note is stopped due to I'm taking a better QFT course. I'm not sure about continuing it or not, thus I just put it here.]

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CHAPTER 1

Fundamental Ideas

This chapter serves as an introduction to basic ideas in QFT, which will be intensively used and discussed in the following chapters. It will not be self contained, yet all preliminaries(free field, path integral, Feymann diagrams) will be found in basic QFT textbooks.

1 Kallen-Lehmann Form of Propagator

This is a topic to prepare us for the LSZ reduction formula. The question is:

- What is a "particle propagating" in a general interacting Lorentz invariant QFT?

To have a concrete understanding of **particle** in interacting theory, we then can know how to QFT to describe the scattering of particles. In free QFT, we know that the 2 point correlation function is interpreted as the amplitude of a particle propagating from one point to another:

$$\langle 0|\mathcal{T}\varphi(x)\varphi(0)|0\rangle \tag{1}$$

However, in general QFT, one may not expect this function to have such interpretation, since the propergator may include propergation of multiparticle states or even particle coming vaccum due to the interaction.

Thus, the role of the KL form is to tell us which part of the 2 point correlation function can be interpreted as the amplitude of a particle propagating, and which part is not.

1.1 KL Form for Scalar Propagator

First we have a look at the massive scalar propagator. We then need to specify our assumptions on the theory, which are:

1. The dynamical fields are scalar fields, and the theory is Poincare invariant.
2. The theory is a sensible quantum theory.

With these minimal assumptions, we can think of the spectrum of the theory can be organized as follow.

1.1.1 Spectrum of a Lorentz Invariant QFT

First, the Poincare invariance of quantum theory means that the theory have a symmetry algebra of Poicare Algebra and the Hilbert space forms a representation of the Poincare Algebra.

A maximally commuting subalgebra of the momentum operator H, P^i, which can be used to label states in the theory. We consider a series of states in this theory with 0 eigenvalues of P^i. We will call them:

$$|\lambda_0\rangle \text{ satisfying } P^i|\lambda_0\rangle = 0 \quad H|\lambda_0\rangle = m_\lambda|\lambda_0\rangle \tag{2}$$

Then we can construct a series of states with nonzero eigenvalues of P^i by applying the Lorentz boost operator K^i on the states $|\lambda_0\rangle$, which will give us a series of states $\big|\lambda_p\big\rangle$:

$$\big|\lambda_p\big\rangle = e^{iK^i\xi_i}|\lambda_0\rangle \tag{3}$$

Due to the fact that $P^\mu P_\mu = m_\lambda^2$ is a Casimir and commute with the Poincare algebra, we know that under the boost operator, the states will satisfy the following eigenvalue equations:

$$P^i\big|\lambda_p\big\rangle = p^i\big|\lambda_p\big\rangle \quad H\big|\lambda_p\big\rangle = E_p(\lambda)\big|\lambda_p\big\rangle \quad E_p(\lambda) = \sqrt{p^2 + m_\lambda^2} \tag{4}$$

Thus in general Hilbert Space of this Lorentz Invariant QFT take form of:

$$\mathcal{H} = \bigoplus_{\lambda,p} \big|\lambda_p\big\rangle \tag{5}$$

> **Remark**
>
> Notice that $\big|\lambda_p\big\rangle$ may not be a single particle states, but just a general momentum and energy eigenstate may have many particle and messy interaction and bounding between them.

1.1.2 Kallen-Lehmann Form of Scalar Propagator

Consider this theory with the above spectrum. We can then write the 2 point correlation function as:

$$\langle\Omega|\mathcal{T}\varphi(x)\varphi(0)|\Omega\rangle \tag{6}$$

Now we insert a complete set of states in the middle, which is:

$$\mathbb{I} = |\Omega\rangle\langle\Omega| + \sum\!\!\!\!\!\!\int_\lambda \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p(\lambda)}\big|\lambda_p\big\rangle\big\langle\lambda_p\big| \tag{7}$$

> **Remark**
>
> notice that here we assume the relativistic normalization of states.

Notice that:

$$\langle\Omega|\varphi(x)|\Omega\rangle = 0 \tag{8}$$

is true for most cases. This is due to the $\mathbb{Z}_2$ symmetry of the scalar theory, if we redefine $\varphi \to -\varphi$ the lagrangian is invariant, yet the correlation function will change sign, thus it must be zero. For highier spin, the fact of being an irreducible representation of lorentz group will also lead to the vanishing of such one point function:

$$\begin{aligned}\langle\Omega|\varphi_a(x)|\Omega\rangle &= \langle\Omega|U(\Lambda)U^\dagger(\Lambda)\varphi_a(x)U(\Lambda)U^{\dagger(\Lambda)}|\Omega\rangle \\ &= D(\Lambda)^b_a\langle\Omega|\varphi_b(\Lambda^{-1}x)|\Omega\rangle\end{aligned} \tag{9}$$

if we take $x=0$, then only when $D(\Lambda)^b_a=\delta^b_a$ we can have nonzero one point function, which is only true for scalar representation. Thus for highier spin representation, the one point function must vanish.

Then we have for $x^0>0$:

$$\langle\Omega|\varphi(x)\varphi(0)|\Omega\rangle = \sum\!\!\!\!\!\!\int_\lambda \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p(\lambda)}e^{-ipx}\langle\Omega|\varphi(x)|\lambda_p\rangle\langle\lambda_p|\varphi(0)|\Omega\rangle \tag{10}$$

Now have a look at the matrix element $\langle\Omega|\varphi(x)|\lambda_p\rangle$. We can translate the field to 0 by the property of $\varphi(x)$ being a lorentz scalar:

$$\langle\Omega|\varphi(x)|\lambda_p\rangle = \langle\Omega|e^{iPx}\varphi(0)e^{-iPx}|\lambda_p\rangle = e^{-ipx}\langle\Omega|\varphi(0)|\lambda_p\rangle \tag{11}$$

Remark

notice the poisson bracket and canonical quantization relation gives us the standard boost generator is $U(a)=e^{iP\cdot a}$, and the relation of:

$$U(a)\varphi(x)U^{\dagger(a)}=\varphi(x+a) \tag{12}$$

and for the lorentz transformation, we have $U(\Lambda)=e^{-\frac{i}{2}J^{\mu\nu}\omega_{\mu\nu}}$ and the relation of:

$$U(\Lambda)\varphi(x)U^\dagger(\Lambda)=\varphi(\Lambda x) \tag{13}$$

By being a scalar, it must satisfy:

$$U(\Lambda)\varphi(0)U^\dagger(\Lambda)=\varphi(0) \tag{14}$$

Thus we can insert a boost operator to the matrix element, which gives us:

$$\langle\Omega|\varphi(0)|\lambda_p\rangle = \langle\Omega|U^\dagger(\Lambda)U(\Lambda)\varphi(0)U^\dagger(\Lambda)U(\Lambda)|\lambda_p\rangle = \langle\Omega|\varphi(0)|\lambda_0\rangle \tag{15}$$

With all these ingrediant we can write the 2 point correlation function as following by making the 3 integral into a 4 integral while the integral of p^0 gives out the energy function exactly, the result is:

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle = \sum_\lambda\int\frac{d^4p}{(2\pi)^4}\frac{i}{p^2-m_\lambda^2+i\epsilon}e^{-ip\cdot(x-y)}|\langle\Omega|\phi(0)|\lambda_0\rangle|^2. \tag{16}$$

Note

The calculation of intergrating out p^0 is a standard trick. the time order of $x^0>y^0$ tells us that the integral contour of p^0 should be closed in the lower half plane, and the only pole is at $p^0=\sqrt{p^2+m_\lambda^2}-i\varepsilon$. Thus this gives us the residue of the integrand at this pole, which is exactly $\frac{1}{2E_p(\lambda)}e^{-ip^0t}$. The other case of $x^0<y^0$ is similar, and the result is the same.

All above deduction is the same for the case of $x^0<y^0$, and takes the exact same form. One may recognize the Feymann propagator within the integral, yet it is in fact a sum

of Feymann propagators with different masses, giving us the Kallen-Lehmann form of the propagator:

Theorem 1.1 **Kallen-Lehmann Form of Propagator**

The 2 point correlation function of a general interacting lorentz invariant QFT can be written as:

$$\langle\Omega|\mathcal{T}\varphi(x)\varphi(y)|\Omega\rangle = \int_0^\infty \frac{dM^2}{2\pi}\rho(M^2)D_F(x-y;M^2), \tag{17}$$

where $D_F(x-y;M^2)$ is the Feymann propagator of a free scalar field and:

$$\rho(M^2) = \sum_\lambda (2\pi)\delta(M^2-m_\lambda^2)|\langle\Omega|\phi(0)|\lambda_0\rangle|^2. \tag{18}$$

For a **typical theory** the spectural density will take the form of:

$$\rho(M^2) = 2\pi Z\delta(M^2-m^2) + \text{continuum} \tag{19}$$

sum over λ gives an isolated point at $m_\lambda = m$ and a continuum of states with $m_\lambda > 2m$. This m is interpreted as the physical mass of a particle and is in general different from the mass parameter in the lagrangian. And Z is the field strength renormalization factor, which is:

$$Z = |\langle\Omega|\phi(0)|\lambda_0\rangle|^2. \tag{20}$$

1.1.3 Interpretation of KL Form

The KL form of the propagator in momentum space behaves as:

$$\int d^4x e^{ipx}\langle\Omega|\mathcal{T}\varphi(x)\varphi(0)|\Omega\rangle = i\frac{Z}{p^2-m^2+i\epsilon} + \text{branch cut terms for } \; p^2 > (2m)^2 \tag{21}$$

we interpret the first term as the amplitude of a single particle propagating. And m as the physical mass of the particle, and Z as the field strength renormalization factor. The second term is the contribution of multiparticle states.

This gives us the slogan that the **Isolated pole** of the correlation function is a **particle** in the interacting theory.

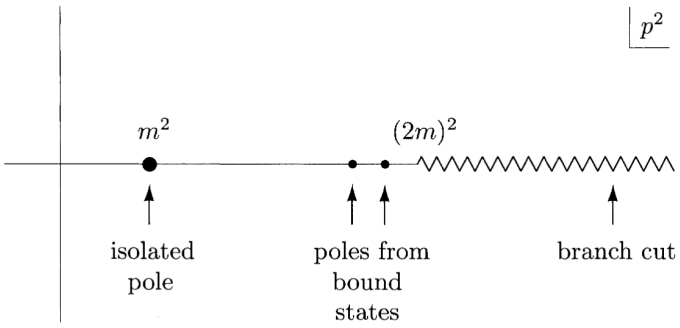


Figure 1: The isolated pole of the correlation function is a particle

1.2 KL Form of Fermion Propagator

For a fermion propagator, the KL form is similar to the scalar case:

$$\int d^4x e^{ipx} \left\langle \Omega \middle| \mathcal{T}\psi(x)\overline{\psi}(0) \middle| \Omega \right\rangle = iZ\frac{\not{p}+m}{p^2-m^2+i\epsilon} + \text{branch cut terms for } \; p^2 > (2m)^2 \quad (22)$$

where the field strength renormalization is defined as:

$$\langle \Omega|\psi(0)|p,s\rangle = \sqrt{Z}u(p,s) \quad (23)$$

Similarly, we interpret this pole as the particle with physical mass m propagating.

1.3 Example: QED Fermion Propagator

Now we look at a concrete example, which is the Fermion propagator in QED, to see if it satisfy this form. We now have tools to deal with interacting theory by doing perturbation, thus we can try to evaluate m and Z perturbatively.

- Leading Order

The leading order of the fermion propagator is just the free propagator:

$$i\frac{\not{p}+m_0}{p^2-m_0^2+i\varepsilon} \quad (24)$$

we'd expect at 0 order $m = m_0$ and $Z = 1$.

> **Remark**
>
> Notice that the above result is a Fourier transformation of the 2 point correlation function, not the Feymann diagram for S matrix, which is not defined yet, but often taught in an early stage.

- First Order

The first order correction due to interaction is the following diagram:

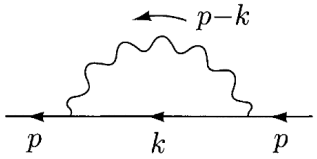


Figure 2: First order correction to the fermion propagator in QED.

> **Note**
>
> To regulate the infrared divergence of the photon, we need a small photon mass μ. With some tricks this won't effect the physical result. This is another important topic.

YL: [better to come back after the electric vertex function example!]

2 LSZ Reduction Formula

For a scattering theory, it is important predict the S-matrix as observable quantity. For QFT, LSZ reduction formula is the one that tells us how to **calculate the S-matrix from the correlation functions**. Then with the tools of Feymann diagrams in correlation function calculation, we can calculate the S matrix using Feymann diagrams.

The idea is quite same as the Kallen-Lehmann form of propagator. We want to know which part of the correlation function can be interpreted as particles interacting and scattering, and which part is not.

3 Optic Theorem

4 Ward-Takahashi Identity

Symmetry plans the fundamental role in QFT, and the Ward-Takahashi identity is a consequence of the symmetry. I'll first talk about the application in electrodynamics, following section 7.4 of [1], then I will talk about the formal structure of the Ward-Takahashi identity following section 9.6 of [1].

4.1 Ward-Takahashi Identity in QED

4.2 Ward-Takahashi Identity in General QFT

CHAPTER 2

Renormalization

Renormalization is the procedure to deal with ultraviolet divergences in quantum field theory. It is a systematic way to cancel these divergences and get finite results for physical quantities.

1 Regualization and Bare Perturbation Theory

Before starting with the systematic renormalization, we need to first regularize the divergences, which is the process to make the divergences controled within some parameters. This is done by introducing a **regularization scheme**. Then we try to deal with the divergences in a naive way, which is called **bare perturbation theory**. This is a cumbersome way to cancel out the divergences, yet it may lead us to understand why the systematic renormalization works.

1.1 Regularization Schemes

There are several regularization schemes, and the most common ones are:

- Pauli-Villars Regularization
- Dimensional Regularization

This section we will intensively study how to calculate loop integrals and do regularization in a standard way. Here are the general steps to folow:

1. Use **Feymann Parameters** to combine the denominators of the propagators in the loop integrals, which will make the loop integrals easier to calculate.

YL: [to be continued see 6.3 of peskin]

1.1.1 Feymann Parameterization

YL: [6.3 in peskin's book!]

1.1.2 Dimensional Regularization

YL: [7.5 in peskin's book!]

1.2 Bare Perturbation Theory

Before going to a systematic way to do renormalization, we need to first try to do it in a more naive and cumbersome way called **bare perturbation theory**.

1.2.1 General Method

Naively we will encountered divergences in observables due to the loops, seems like the theory is ill defined. However, in fact its is our understanding of the theory is incomplete, here is a modern understanding:

- We have infinity due to the fact that we are using the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, ...$ to express the S Matrix, thus the coefficients turns out to be infinite.
- We have to define some physical quantities (and argue why they are physical) and think of them as the result of stuffs we really measure in experiments, such as the physical mass m, the physical coupling constant λ ...
- Now we can express the S Matrix in terms of the physical quantities, turns out that the coefficients are finite, and we can get finite results for physical quantities.

To do this, we follow the following steps:

1. **Regularize the divergences in Observable** (eg. S Matrix), and express the Observable in terms of the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, ...$, with infinite divergences.
2. **Define the physical quantities**, such as the physical mass m, the physical coupling constant λ and calculate them perturbatively in terms of the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, ...$ to some orders.
3. **Express the Observable in terms of the physical quantities**, and we will find that the coefficients are finite, and we can get finite results for physical quantities.

> **Note**
>
> It is important to have all steps in the same order of perturbation. For tree level, there are no loops and divergence, thus we can directly express bare quantity as physical quantities, if we fix the order of perturbation to be tree level.

1.3 Example: Electric Vertex Function

Lets use the mathematical tools developed in the previous sections to calculate some examples of loop integrals and do regularization, which will be useful for the following sections on renormalization.

The electric vertex function is the most basic example of renormalization, which is the one-loop correction to the electric vertex function, given in section 6.3 and 7.2, 7.4 of [1].

1.3.1 Electric Vertex Function

1.3.2 Evaluation of Electric Vertex Function

1.3.3 Canceling Ultraviolet Divergence

1.3.4 Divergence Cancellation in all orders

There is a problem of can field strength renormalization cancel divergence to all orders? This is enforced by the Ward-Takahashi identity, which is a consequence of the gauge symmetry.

1.4 Example: Vaccum Polarization

This is another example from Peskin and Schroeder [1], which is the one-loop correction to the photon propagator, given in section 7.5.

2 Renomalizability

Above sections we show that some of the divergence in a theory may be canceled out using a renomalizing technique. However, for a sensible theory, we need to make sure that all the divergences can be canceled out when expressing the physical obserables using physical values. Thus, before developing a systematic way to do renormalization, we need to first classify theories based on whether they can be renomalized or not.

To answer this question, we have 2 subquestions:

1. How to count the number of divergent amplitudes in a theory?
2. What kind of divergence is controllable? (ie. can be canceled out by renormalization)

2.1 How to Count Divergences?

We have to answer the first question first.

- How to count the number of divergent amplitudes in a theory?

We introduce the concept of **superficial degree of divergence**. To make things easy, we take a φ^n theory in d dimensional spacetime as an example:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 - \frac{1}{2}m^2\varphi^2 - \frac{\lambda}{n!}\varphi^n \tag{25}$$

Now consider a Feymann Diagram with:

$$\begin{aligned} &N : \text{number of external lines} \\ &P : \text{number of propagators} \\ &V : \text{number of vertices} \\ &L : \text{number of loops} \end{aligned} \tag{26}$$

We can define a quantiy called **superficial degree of divergence** as:

Definition 2.1 **Superficial Degree of Divergence**

The superficial degree of divergence of a Feymann Diagram is defined as:

$$D = dL - 2P \tag{27}$$

where L is the number of loops and P is the number of propagators in the Feymann Diagram.

Why does this quantity characterize the divergence of the Feymann Diagram? We can have a loop at the Feymann Rules:

- Naively, each Propagator contributes a factor of $1/k^2$ to the amplitude, which gives a factor of -2 to the degree of divergence.

$$D_F \sim \frac{1}{k^2 + m^2} \tag{28}$$

- Each loop contains a momentum that needs to be integrated over, which gives a factor of;

$$\int \frac{d^d k}{(2\pi)^d} \tag{29}$$

Thus the total contribution to the degree of divergence is:

$$D = dL - 2P \tag{30}$$

There is also an expression using numbers of vertices and external lines:

$$\begin{aligned} L = P - V + 1 \quad nV = 2P + N \\ \Rightarrow D = d - \frac{d-2}{2}N - V\left(d - n\frac{d-2}{2}\right) \end{aligned} \tag{31}$$

However, we know that this is really just a simplified version of the true divergence. Because there are some special cases that have different divergence than the superficial degree of divergence.

Commonly these special cases are given in the following list (this list is exactly true for QED, that this list contains all the special cases for QED):

1. **Sub-Diagram Divergence**: A Feymann Diagram may contain a sub-diagram that has a divergence, which may gives larger divergence than the superficial degree of divergence.
2. **Symmetry**: Some Feymann Diagrams may have some symmetry that ensures some divergent diagrams canceled out in some amplitudes, which makes the true divergence smaller than the superficial degree of divergence.
3. **Trivial diagram**: trivial diagram with no loops and no propagators, which has a degree of divergence of $D = 0$ but is actually finite.

We can argue that these special cases have no effect on our analysis of divergence and renormalizability:

1. For the first case, we can just consider all amputated, one particle irreducible basic amplitudes, and make sure they are controlable. If one highier diagram is superficially converge, yet truely diverge, the only possibility is to have a sub-diagram in the cases we discussed, which is controlable.
2. For the second case, we would like to see this.
3. For the third case, just ignore it, no one cares about it.

Thus, we'd like to say that the superficial degree of divergence is a good way to characterize the divergence of the Feymann Diagram. There are conter examples, especially for more complicated theories like Yang-Mills theory, this will be discussed in more advanced sections.

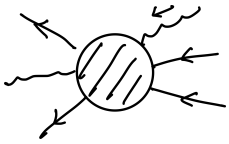
2.1.0.1 What kind of Divergence is Controllable?

The second question is what kind of divergence is controllable, which means that we can cancel out the divergence by renormalization. The answer is that:

- **Super Renormalizable**: Theories with only a finite number of superficially divergent diagrams: which means that 1. only a finite number of amplitudes diverges 2. they only diverge to a finite order.
- **Renormalizable**: Theories with an infinite number of superficial divergent amplitudes: which means that 1. an infinite number of amplitudes diverges 2. they may diverge in any order.
- **Nonrenormalizable**: Theories with an infinite number of superficial divergent amplitudes.

Remark

Here for Amplitude, we means general stuffs like this, which contains summing over all Feymann Diagrams to all orders.



If an amplitude diverges, it may have two cases: either Feymann Diagrams to all orders have divergence or only a Feymann Diagram of some finite order diverges. This difference separates the superrenormalizable theories and renormalizable theories.

As the name suggests, only a renormalizable theory is renormalizable.

- Why is this the case??

A basic analysis is that ther are only finite number of parameters we can use as physical quanitities. If there are too many infinities, they may not be adiquate enough for cancelation. And if we have finite numbers of amplitudes that diverges superficially, we can have finite physical quantities to plug in and cancel out the divergences. We will see examples to justify this claim.

Thus we can have a partically systematic way to see whether a theory is renormalizable or not:

- **Step 1:** try to list out all superficially divergent, amputated, one particle irreducible basic amplitudes.
- **Step 2:** check whether there are truely divergent or not; there are more divergent amplitudes beyond superficial expectations.

We will finally see examples of this procedure.

2.2 Mass Dimension and Renormalizability

2.2.1 Divergence and Mass Dimension

There is also a more straightforward way of seeing divergence of a Feymann Diagram. We know that an amplitude should be of the same dimension.

consider a n external leg diagram in the φ^n theory:

- The tree level diagram:

$$i\mathcal{M} = -i\lambda \tag{32}$$

By dimensional analysis, we have:

$$[\lambda] = d - n\frac{d-2}{2} \tag{33}$$

- Consider a loop diagram of this n external leg amplitude, which has V vertices, the amplitude is:

$$i\mathcal{M} \sim \lambda^V \Lambda^D \tag{34}$$

where Λ is the cutoff scale, and D is the superficial degree of divergence.

- By dimensional analysis, we have:

$$[\lambda^V \Lambda^D] = V\left(d - n\frac{d-2}{2}\right) + D = d - n\frac{d-2}{2} \tag{35}$$

inverse the equation, we have:

$$D = d - n\frac{d-2}{2} - V\left(d - n\frac{d-2}{2}\right) \tag{36}$$

which is exactly the same as the superficial degree of divergence we have defined before.

2.2.2 Renormalizability from Mass Dimension

These relations in fact is much physical than just a mathematical relation. We then can see the mass dimension of the coupling constant gives us a direct way to see whether a theory is renormalizable or not:

- **Super Renormalizable**: Theories with coupling constants of positive mass dimension.
- **Renormalizable**: Theories with coupling constants of zero mass dimension.
- **Non Renormalizable**: Theories with coupling constants of negative mass dimension.

In fact this can also be justified in a EFT view.

2.3 Example: Renormalizability of QED

YL: [10.1 in peskin's book!]

Lets apply this two steps to QED:

- **Step 1:** try to list out all superficially divergent, amputated, one particle irreducible basic amplitudes.
- **Step 2:** check whether there are truely divergent or not; there are more divergent amplitudes beyond superficial expectations.

The list of all superficially divergent, amputated, one particle irreducible basic amplitudes in QED is given in the following table:

3 Renormalized Perturbation Theory

YL: [here shall follow peskin and use phy4 theory as an example]

3.1 General Method (Example: φ^4 Theory)

Renormalization Condition is in fact defining what the physical quantities are in this theory.

3.2 Example: Renormalization of QED

Bibliography

[1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*. Reading, Mass: Addison-Wesley Pub. Co, 1995.